

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA
KAPREKAR CONTEST – FINAL – SUB JUNIOR
 Classes VII & VIII **2017**



Instructions:

1. Answer as many questions as possible.
2. Elegant and novel solutions will get extra credits.
3. Diagrams and explanations should be given wherever necessary.
4. Fill in FACE SLIP and your rough working should be in the answer book.
5. Maximum time allowed is THREE hours.
6. All questions carry equal marks.

1. (a) Find all three digit numbers in which any two adjacent digits differ by 3.
 (b) There are 5 cards. Five positive integers (may be different or equal) are written on these cards, one on each card. Abhiram finds the sum of the numbers on every pair of cards. He obtains only three different totals 57, 70, 83. Find the largest integer written on a card.
2. (a) ABC is a triangle in which $AB = 24, BC = 10$ and $CA = 26$. P is a point inside the triangle. Perpendiculars are drawn to BC, AB and AC . Length of these perpendiculars respectively are x, y and z . Find the numerical value of $5x + 12y + 13z$.
 (b) If

$$x^2(y+z) = a^2, \quad y^2(z+x) = b^2, \quad z^2(x+y) = c^2, \quad xyz = abc$$

prove that $a^2 + b^2 + c^2 + 2abc = 1$

3. If

$$X = \frac{a^2 - (2b - 3c)^2}{(3c + a)^2 - 4b^2} + \frac{4b^2 - (3c - a)^2}{(a + 2b)^2 - 9c^2} + \frac{9c^2 - (a - 2b)^2}{(2b + 3c)^2 - a^2}$$

$$Y = \frac{9y^2 - (4z - 2x)^2}{(2x + 3y)^2 - 16z^2} + \frac{16z^2 - (2x - 3y)^2}{(3y + 4z)^2 - 4x^2} + \frac{4x^2 - (3y - 4z)^2}{(4z + 2x)^2 - 9y^2}$$

find $2017(X + Y)$

4. The sum of the ages of a man and his wife is six times the sum of the ages of their children. Two years ago the sum of their ages was ten times the sum of the ages of their children. Six years hence the sum of their ages will be three times the sum of the ages of their children. How many children do they have?
5. (a) a, b, c are three natural numbers such that $a \times b \times c = 27846$. If $\frac{a}{6} = b + 4 = c - 4$, find $a + b + c$.
 (b) $ABCDEFGH$ is a regular octagon with side length equal to a . Find the area of the trapezium $ABDG$.

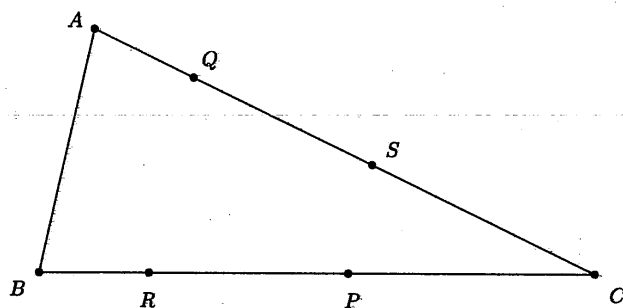
6. (a) If a, b, c are positive real numbers such that no two of them are equal, show that

$$a(a-b)(a-c) + b(b-c)(b-a) + c(c-a)(c-b)$$

is always positive.

- (b) In the figure below, P, Q, R, S are points on the sides of the triangle ABC such that

$$CP = PQ = QB = BA = AR = RS = SC$$



Find the $\angle C$.